

Physics-Informed Multilayer Perceptron for Predicting the Performance Ratio of Photovoltaic Systems

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Abstract—The accurate estimation of the performance ratio (PR) is an important aspect in the design of effective photovoltaic (PV) systems. The classic approach is based on simulation tools that model solar irradiation, system components, and expected losses. Different configurations are defined by varying parameters such as module orientation and tilt angle, and the resulting PR values are compared. To simplify the design, this paper introduces a neural model that directly provides the best PR for a specific configuration. In line with the idea introduced by Physically Informed Neural Networks (PINNs), this neural model is forced to satisfy additional constraints specific to the physical system, namely, providing daily solar irradiation (H_{poa}) within natural limits. These constraints are addressed by additional terms in the loss function. To build the training and testing datasets, a comprehensive simulation scenario was implemented to extract different thermo-electrical characteristics. This was followed by a class distribution balancing and computation of additional features. A comparative experimental analysis demonstrated that the model can learn to satisfy the imposed constraints, and a penalty factor ($\lambda = 0.01$) maintains high accuracy as well.

Index Terms—Photovoltaic Systems, Machine Learning, Neural Networks

I. INTRODUCTION

In the context of the global energy transition, the efficiency of photovoltaic (PV) power plants represents a priority. The fundamental parameter used to evaluate the “health” of a PV system is the performance ratio (PR). This metric quantifies the efficiency with which solar energy is converted into useful electrical energy, independent of the plant’s geographical location or orientation. Consequently, the accurate prediction of this parameter is essential for modern automated monitoring systems.

In recent literature, approaches based on Machine Learning (ML) algorithms, such as Multilayer Perceptrons, Long Short-Term Memory networks, and Gated Recurrent Units, have become the standard for the estimation of PR due to their

ability to model complex, non-linear correlations between electrical and meteorological data [7], [8].

However, classical algorithms suffer from a major drawback, they operate as “black boxes” entirely ignoring the laws of physics. They optimise a purely mathematical error metric (e.g., Mean Squared Error - MSE), so due to a faulty sensor or unexpected weather conditions, it can output electrical or thermodynamically impossible predictions [6].

To overcome the limitations of the classical models and provide a robust solution for the industrial environment, this paper proposes the implementation of a neural network that satisfies constraints imposed by the physical system [5]. By anchoring the laws of energy conservation directly into the network’s optimization process, the model is forced to respect the thermodynamic constraints.

The main contributions of this research are:

- **Feature Engineering:** Extracting six physical indicators of thermo-electrical stress (e.g., thermal efficiency, DC/AC overload ratio) from a set of 10 raw parameters derived from PV modeling simulations, carried out in SAM (System Advisor Model) [2] and specific meteorological datasets [4]. This facilitates the prediction of the performance ratio by the neural network.
- **Balancing the Operational Space:** Implementing oversampling techniques to ensure uniform distribution of operational classes, eliminating the algorithm’s bias towards the majority states of normal operation.
- **Development of an MLP satisfying physical constraints:** Formulating a customized loss function that penalizes the network whenever the predicted PR indicator implies a specific daily solar irradiation (H_{poa}) located outside a valid physical range.
- **Sensitivity Analysis and Constraint Optimization:** Conducting a Grid Search evaluation to optimize the

physical penalty factor (λ) and the irradiation limits, demonstrating how the neural model succeeds in rejecting incorrect, unrealistic cases.

II. DATA PREPROCESSING AND METHODOLOGY

A. Data Acquisition and Feature Engineering

The foundation of this research is a highly filtered dataset generated through iterative simulations using the National Renewable Energy Laboratory's (NREL) System Advisor Model (SAM) [1], [2]. Meteorological boundary conditions were defined using localized irradiance and temperature profiles from the National Solar Radiation Database (NSRDB) for Iași, Romania [4].

The raw dataset (Table I) comprises 10 primary features, including electrical parameters (e.g., V_{mpp} , I_{mpp} , V_{dc_max}) and hardware configurations (the number of modules and parallel strings). The parameters extracted for each simulated configuration include:

- **Hardware Configuration Variables**

- **Modules per string ($N_{modules}$):** The number of PV panels connected in series to form a single string, directly dictating the overall voltage.
- **Parallel Strings ($N_{strings}$):** The total number of identical strings connected in parallel to the inverter, dictating the total input current.
- **Total DC Capacity (P_{dc_max}):** The total installed nominal DC power of the photovoltaic array (expressed in kW).

- **Module Electrical Characteristics**

- **Maximum Power Point Voltage (V_{mpp}):** The voltage at which a single module produces its maximum power under Standard Test Conditions (STC).
- **Open-Circuit Voltage (V_{oc}):** The maximum voltage available from a module when there is no current flowing.
- **Maximum Power Point Current (I_{mpp}):** The current generated by the module at its peak power output.
- **Temperature Coefficient (γ):** The rate at which the module's power output degrades for every degree Celsius above $25^\circ C$ (expressed in $\%/^\circ C$).

- **Inverter Electrical Characteristics**

- **Maximum DC Voltage (V_{dc_max}):** The absolute maximum DC input voltage the inverter can safely accept without hardware damage.
- **Minimum MPPT Voltage (V_{mppt_min}):** The lower boundary of the inverter's voltage window required to successfully track the maximum power point.
- **Maximum Input Current (I_{inv_max}):** The maximum total DC current the inverter is capable of processing.
- **Maximum AC Power (P_{ac_max}):** The rated nominal active AC power output capacity of the inverter (expressed in W).

- **Environmental and Output Variables**

- **Total AC Energy (E_{out}):** The final annual energy yield delivered to the grid.
- **Performance Ratio (PR):** The target variable, representing the overall efficiency of the system.

To enhance the neural network's ability to map the non-linear relationship between hardware stress and performance degradation, six more features were synthesized.

- **Thermal Efficiency Coefficient ($\eta_{thermal}$):** Accounts for performance degradation due to ambient temperature (T_{amb}) deviations from Standard Test Conditions ($25^\circ C$).

$$\eta_{thermal} = 1 + \left(\frac{\gamma}{100} \right) \cdot (T_{amb} - 25) \quad (1)$$

where γ is the manufacturer-specified temperature coefficient.

- **Voltage Stress Ratio (V_{ratio}):** Represents the proximity of the operating string voltage to the inverter's maximum permissible DC limit.

$$V_{ratio} = \frac{N_{modules} \cdot V_{mpp}}{V_{dc_max}} \quad (2)$$

- **Current Stress Ratio (I_{ratio}):** Describes the total input current relative to the inverter's maximum rated capacity

$$I_{ratio} = \frac{N_{strings} \cdot I_{mpp}}{I_{inv_max}} \quad (3)$$

- **Design DC/AC Ratio:** The ratio of installed DC power to the rated AC output capacity, a critical metric for clipping probability [3].

$$DC/AC = \frac{P_{dc_max} \cdot 1000}{P_{ac_max}} \quad (4)$$

- **Relative Module Efficiency (η_{rel_mod}):** The instantaneous electrical efficiency of the module relative to its nominal Standard Test Conditions rating.

$$P_{mod_STC_kW} = \frac{P_{dc_max}}{N_{modules} \cdot N_{strings}} \quad (5)$$

$$\eta_{rel_mod} = \frac{V_{mpp} \cdot I_{mpp} / 1000}{P_{mod_STC_kW}} \quad (6)$$

where ($P_{mod_STC_kW}$) is the nominal STC power rating.

- **Voltage Safety Margin (V_{margin}):** The absolute difference between the inverter's maximum DC limit and the actual string voltage.

$$V_{margin} = V_{dc_max} - (N_{modules} \cdot V_{mpp}) \quad (7)$$

B. Dataset Characterization and Class Balancing

The PR was utilized as the primary target variable [11]. To facilitate robust diagnostic capabilities, the dataset was categorized into four distinct operational zones based on established PR thresholds:

- **Optimal ($0.78 \leq PR \leq 0.92$):** System operating within nominal design parameters.
- **Overclipping ($0.60 \leq PR < 0.78$):** Performance limited by inverter saturation due to high irradiance.

TABLE I
SAMPLE OF THE RAW PHOTOVOLTAIC DATASET EXTRACTED FROM SAM SIMULATIONS

Modules / String	Parallel Strings	V_{mpp} (V)	V_{oc} (V)	Temp. Coeff. (%/°C)	I_{mpp} (A)	V_{dc_max} (V)	V_{mppt_min} (V)	I_{inv_max} (A)	P_{ac_max} (W)	H_{poa} (kWh/m ²)	T_{amb} (°C)	E_{out} (kWh)	P_{dc_max} (kW)	PR	System Status
2.0	9.0	176.1	216.1	-0.24	2.3	480	200.0	17.60	6149.0	3.53	10.8	8742.18	7.20	0.829	Optimal
2.0	10.0	176.1	216.1	-0.24	2.3	425	240.0	25.00	8980.0	3.53	10.8	9690.65	7.99	0.827	Optimal
2.0	6.0	176.1	216.1	-0.24	2.3	450	75.6	15.15	3810.0	3.53	10.8	5810.79	4.80	0.826	Optimal
3.0	10.0	176.1	216.1	-0.24	2.3	480	360.0	29.27	11390.0	3.53	10.8	14254.00	11.99	0.811	Optimal
5.0	10.0	176.1	216.1	-0.24	2.3	950	790.0	39.84	33300.0	3.53	10.8	24882.90	19.99	0.849	Optimal

- **Failure** ($PR < 0.30$): Severe hardware malfunction or total system shutdown.
- **Gray Zone** ($0.30 \leq PR < 0.60$): Transitional or physically anomalous states.

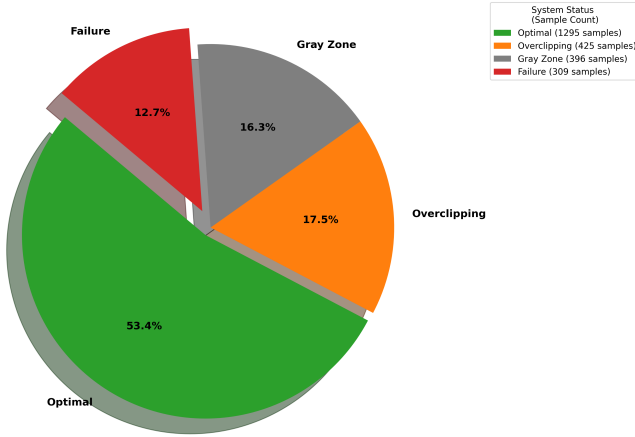


Fig. 1. Distribution of Operational States based on Performance Ratio (PR).

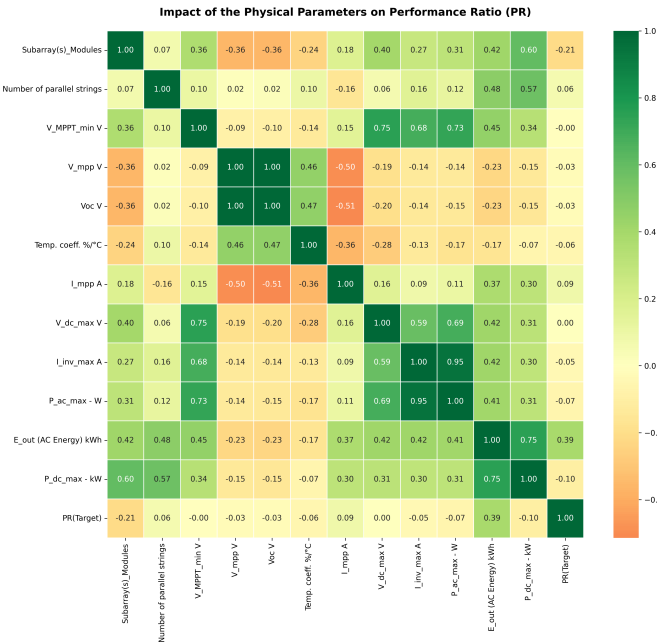


Fig. 2. Correlation Matrix of Raw Photovoltaic Parameters and the Performance Ratio (PR).

Initial exploratory analysis revealed a significant class imbalance, with "Optimal" samples overwhelmingly dominating the dataset. To prevent the ML model from developing a statistical bias toward normal operation, a uniform oversampling technique was applied. This process involved replicating minority class samples (specifically "Failure" and "overclipping") until all classes reached the same number of samples, so the neural network's architecture learns the physical boundaries of the system failures with high precision, rather than ignoring them as statistical outliers [9].

As illustrated in the correlation matrix, the direct linear relationships between the raw physical parameters and the target, PR, are not significant. Among the raw variables, the system's sizing parameters (such as the number of modules and strings) and the inverter's electrical constraints (V_{dc_max} and I_{inv_max}) are the most impactful, as they dictate the thresholds where performance drops due to overclipping. These specific raw hardware limits will serve as the mathematical anchors for the custom physical loss function. By feeding these constraints into the neural network, the model will learn to autonomously distinguish between a natural efficiency drop and an actual hardware anomaly.

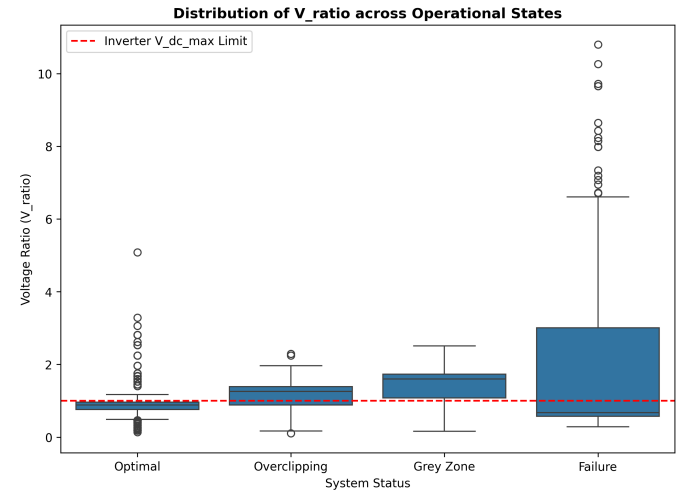


Fig. 3. Distribution of the engineered Voltage Stress Ratio (V_{ratio}) across the established operational states. The dashed line represents the physical maximum DC voltage limit of the inverter.

The distributional analysis of the engineered Voltage Stress Ratio (V_{ratio}) reveals the strict physical boundaries governing the system's operational states. As previously defined in equation (2), this indicator quantifies the proximity of the operating string voltage to the inverter's absolute maximum DC limit.

As shown in Fig. 3, the boxplot demonstrates a definitive threshold: samples reaching or exceeding the inverter limit ($V_{ratio} \geq 1.0$) fall entirely outside the optimal zone, indicating severe thermo-electrical stress [3].

III. NEURAL MODEL NETWORK ARCHITECTURE

A. Baseline Statistical Model

To establish a benchmark for the performance and to highlight the necessity of physical constraints, a standard Multi-Layer Perceptron (MLP) was initially developed. This baseline model operates as a classic feed-forward Artificial Neural Network (ANN). The architecture consists of an input layer receiving the preprocessed meteorological data and engineered electrical features, followed by three hidden layers configured with a 150-100-50 neuron topology. The Rectified Linear Unit (ReLU) activation function was applied across all hidden layers to capture complex non-linear relationships, terminating in a single linear output neuron dedicated to predicting the Performance Ratio ($\widehat{PR}$).

During the training phase, this purely data-driven model relies exclusively on minimizing the Mean Squared Error (MSE) between the predicted and actual PR values. The objective function is defined strictly as:

$$\mathcal{L}_{baseline} = \frac{1}{N} \sum_{i=1}^N \left(PR_{true}^{(i)} - \widehat{PR}^{(i)} \right)^2 \quad (8)$$

where N represents the batch size, $\widehat{PR}^{(i)}$ is the model's prediction, and $PR_{true}^{(i)}$ is the measured target.

While the standard MLP demonstrates strong curve-fitting capabilities under nominal environmental conditions, its fundamental limitation lies in its purely statistical "black-box" nature. Because the loss function ($\mathcal{L}_{baseline}$) optimization works without any inherent awareness of photovoltaic thermodynamics, the model remains completely blind to the hardware constraints established during the feature engineering phase. This critical vulnerability necessitates the transition from a standard statistical algorithm to a physics-informed architecture capable of bounding predictions within the bounds of reality.

B. Architecture Variations and Simulation Results

To systematically evaluate the generalization capacity of the baseline model, six distinct MLP configurations were conducted. A constant network topology was maintained across all experiments to isolate the impact of input data quality and class balancing techniques. Performance was quantified using Coefficient of Determination (R^2) and Mean Squared Error (MSE).

Normal Imbalanced Set:

- **Architecture 1 Raw Data:** This baseline configuration utilized the 10 primary raw physical variables. Although the model achieved a high training R^2 score of **0.9529**, its performance on the test set dropped significantly to an R^2 of **0.8241** (with an MSE of **0.00918**). This discrepancy

indicates a limited capacity to manage variance and suggests overfitting [8].

- **Architecture 2 Calculated Indicators:** In this experiment, the input was restricted exclusively to the 6 thermo-electrical indicators (e.g., V_{ratio} , I_{ratio}) [3]. The results yielded the weakest performance of the study, recording a test R^2 of **0.7244** and a corrected test MSE of **0.01439**. This simulation demonstrated that while calculated indicators are valuable for establishing physical boundaries, they lack the granular information required for precise, stand-alone regression tasks.
- **Architecture 3 Combined Data:** By fusing the 10 raw variables with the 6 calculated indicators (totaling 16 input parameters), the model registered a significant improvement, achieving a test R^2 of **0.8705** and a test MSE of **0.00675**. This synergistic effect confirms that supplying the network with both foundational raw data and explicit knowledge of hardware limits enhances its ability to map stress zones accurately [7].

Balanced Set:

- **Architecture 4 Raw Data:** Applying an oversampling class balancing technique to the raw dataset generated a major qualitative leap in predictive power. The model achieved an R^2 of **0.9652** on the test set, drastically reducing the predictive error (MSE = **0.00248**). This results highlights a critical reality in PV monitoring.
- **Architecture 5 Combined Data:** Even when trained on a balanced dataset, the exclusive use of calculated indicators exhibited inherent limitations, achieving a test R^2 of **0.8456**. Furthermore, the elevated test MSE of **0.10998** suggests that while the model captures the general performance trend, it struggles to predict fine grained PR values.
- **Architecture 6 Combined Data:** This architecture represents the best configuration identified during the statistical baseline testing. The integration of all 16 parameters within a balanced dataset yielded the highest test precision (R^2 = **0.9660**) and the lowest error margin (MSE = **0.00242**).

To fully appreciate why data balancing is a mandatory prerequisite before implementing advanced physical constraints, a direct visual comparison between Architecture 1 (**imbalanced**) and Architecture 4 (**balanced**) was conducted. Both architectures utilized the exact same 10 raw input parameters, effectively isolating the impact of class distribution.

As illustrated in Fig. 4, Architecture 1 exhibits severe predictive scattering because its gradient descent optimization heavily favored the abundant optimal scenarios over critical minority states like failure or overclipping. To rectify this inherent bias, Architecture 4 applied an oversampling technique to synthetically duplicate these minority classes, thereby enforcing equal statistical weight across all operational states during backpropagation [9]. As depicted in Fig. 5, this methodology drastically tightened the predictions along the regression axis, visually confirming that addressing statistical

asymmetry is a mandatory prerequisite for stable regression prior to integrating the thermodynamic Double Bound.

Consequently, the integration of the Double Bound is strictly necessary to impose rigid physical ceilings, effectively transforming a statistically optimized network into a physically diagnostic tool.

IV. DEVELOPMENT OF AN MLP SATISFYING PHYSICAL CONSTRAINTS

A. Derivation of the Thermodynamic Boundaries

To effectively constrain the neural network, its dimensionless output must first be mathematically mapped to a tangible thermodynamic quantity. While the standard MLP directly predicts the $\widehat{PR}$, this metric alone lacks explicit physical boundaries, making it difficult to mathematically penalize impossible predictions. The derivation begins with the fundamental definition of the PR [11]:

$$PR = \frac{E_{out}}{365 \cdot P_{dc_max} \cdot H_{poa}} \quad (9)$$

where 365 extrapolates the daily specific yield to an annual baseline and E_{out} is the measured active AC energy.

To utilize this relationship as a physical constrain during model optimization, the equation is algebraically rearranged to isolate the specific solar irradiation. During the neural network's training phase, the true PR is replaced by the model's active prediction ($\widehat{PR}$). Consequently, the network's output is dynamically reverse calculated at each epoch to determine the exact specific irradiation ($\hat{H}_{poa}$), required to physically achieve that predicted performance:

$$\hat{H}_{poa} = \frac{E_{out}}{365 \cdot P_{dc_max} \cdot \widehat{PR}} \quad (10)$$

For this specific implementation, a **Double Bound** envelope was determined by two limits:

- **The Lower Bound (H_{poa}^{lower}):** Represents the minimum threshold of solar irradiation required to overcome the inverter's wake-up voltage. Predictions implying irradiation below this threshold during active generation states are physically invalid.
- **The Upper Bound (H_{poa}^{upper}):** Represents the absolute maximum theoretical irradiation [13].

If the neural network outputs a $\widehat{PR}$ that implies an irradiation level exceeding H_{poa}^{upper} or falling below H_{poa}^{lower} , the prediction is definitively classified as an algorithmic hallucination. These predefined limits form the corridor that serves as the mathematical foundation for the custom loss function.

B. Formulation of the Custom Loss Function

Standard neural networks optimize their weights by minimizing a purely statistical loss function (MSE). While the MSE effectively captures the data variance ($\mathcal{L}_{data}$), it remains blind to the physical reality. To construct the custom loss function, a hybrid objective function was developed, penalizing

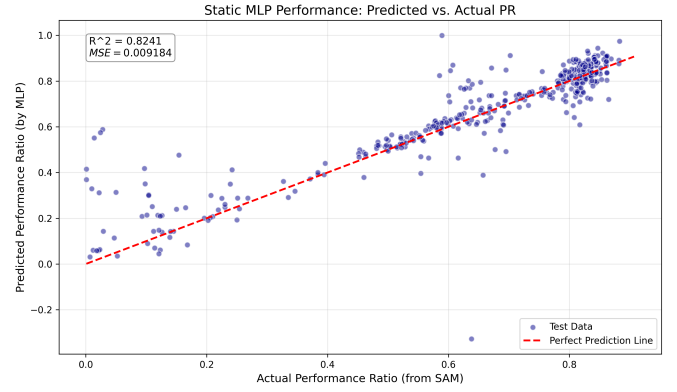


Fig. 4. Regression analysis of Architecture 1 trained on the raw, imbalanced dataset. The wider dispersion from the ideal $y = x$ reference line indicates high predictive variance during minority operational states.

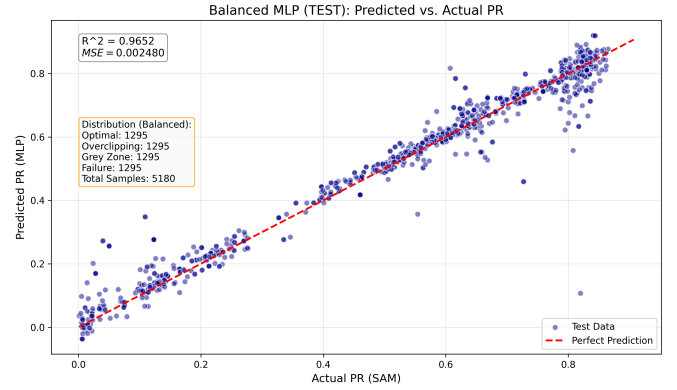


Fig. 5. Regression analysis of Architecture 4 trained on the oversampled dataset. The synthetic equalization of classes significantly tightens the predictions along the regression axis.

the network whenever its output violates the predefined H_{poa} boundaries [5].

Mathematically defined as $f(x) = \max(0, x)$, the ReLU function is exceptionally well-suited for boundary enforcement because it outputs zero for any negative input and scales linearly for any positive input. By passing the boundary conditions through the ReLU function, the physical loss component ($\mathcal{L}_{physics}$) is formulated as:

$$\mathcal{L}_{physics} = \text{ReLU}(H_{poa}^{lower} - \hat{H}_{poa}) + \text{ReLU}(\hat{H}_{poa} - H_{poa}^{upper}) \quad (11)$$

This formulation creates a highly specific punitive mechanism:

- **Lower Bound Violation:** If the predicted irradiation ($\hat{H}_{poa}$) drops below the minimum threshold (H_{poa}^{lower}), the term $(H_{poa}^{lower} - \hat{H}_{poa})$ becomes positive. The first ReLU activates, generating a penalty proportional to the magnitude of the violation.
- **Upper Bound Violation:** If the prediction exceeds the absolute maximum value (H_{poa}^{upper}), the term $(\hat{H}_{poa} - H_{poa}^{upper})$ becomes positive. The second ReLU activates, heavily penalizing the thermodynamic impossibility.
- **Physical Compliance:** If the prediction falls strictly within the established thermodynamic corridor

($H_{poa}^{lower} \leq \hat{H}_{poa} \leq H_{poa}^{upper}$), both arguments evaluate to negative values. Consequently, both ReLU functions output exactly zero, meaning no physical penalty is applied, and the network is optimized solely on its statistical accuracy.

The total loss function ($\mathcal{L}_{total}$) optimized during the model training phase is as follows:

$$\mathcal{L}_{total} = \frac{1}{N} \sum_{i=1}^N \left(PR_{true}^{(i)} - \widehat{PR}^{(i)} \right)^2 + \lambda \cdot \mathcal{L}_{physics} \quad (12)$$

where N is the batch size and λ is a critical, dimensionless hyperparameter known as the penalty factor. The inclusion of λ is mandatory to balance the magnitude of the MSE.

C. Calibration of the Penalty Factor (λ)

This hyperparameter dictates the training equilibrium between statistical accuracy (minimizing MSE) and thermodynamic compliance (minimizing $\mathcal{L}_{physics}$). Selecting the appropriate value for λ requires a delicate balance.

Through an empirical grid search evaluating various values, it was observed that setting a disproportionately large penalty factor severely degraded the model's performance [10]. When λ is too large, the physical penalty term overwhelms the objective function. Conversely, if λ is too small (e.g., $< 10^{-3}$), the penalty generated during a boundary violation is insufficient to alter the network's weights.

To resolve this trade-off, the penalty factor was optimally calibrated to $\lambda = 0.01$. This specific value establishes a perfect functional equilibrium. At $\lambda = 0.01$, the network is permitted to smoothly optimize its weights to capture fine-grained data patterns (minimizing the MSE) during nominal operation. However, the moment a prediction breaches the H_{poa}^{lower} or H_{poa}^{upper} thresholds, the ReLU activation ensures the scaled penalty ($\lambda \cdot \mathcal{L}_{physics}$) is sufficiently rigid to instantly correct the trajectory of the gradient descent, pulling the prediction back into the realm of thermodynamic reality without destabilizing the overall learning process.

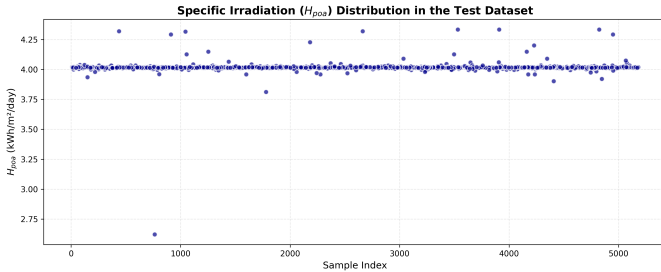


Fig. 6. Distribution of the derived Specific Plane-of-Array Irradiation (H_{poa}) across the standardized test dataset. This natural variance represents the explicit physical domain within which the network's predictions are constrained.

The derived specific irradiation (H_{poa}) was mapped across the standardized test dataset. Because all architectural variations were evaluated using a fixed random seed, this dataset serves as the universal baseline for assessing predictive stability. As illustrated in Fig. 6, the H_{poa} values exhibit

significant natural variance, in the proposed neural model, every single abstract PR prediction made on this dataset is dynamically transformed into this explicit physical domain. By continuously evaluating the outputs against the rigid H_{poa}^{lower} and H_{poa}^{upper} thresholds, the custom loss function guarantees that the network's gradient descent remains strictly anchored to meteorological reality. Ultimately, this mathematical formulation successfully bridges the gap between purely statistical pattern recognition and absolute thermodynamic compliance.

V. IMPACT OF THERMODYNAMIC CONSTRAINTS ON PREDICTIVE ACCURACY

A. Performance with Partial Constraints (Single Lower Bound)

To validate the efficiency of the custom loss function, the thermodynamic constraints were introduced sequentially. The first phase of integration involved applying only the lower threshold (H_{poa}^{lower}) to the optimal baseline architecture.

By modifying the objective function to include only the lower boundary penalty: $\mathcal{L}_{partial} = \text{MSE} + \lambda \cdot \text{ReLU}(\hat{H}_{poa} - H_{poa}^{lower})$, the network was forced to recognize the physical minimums of the systems. This isolated success at the lower boundary, coupled with continued failure at the upper boundary, definitively proved that a partial physical constraint is insufficient for industrial PV monitoring, necessitating the deployment of the complete Double Bound envelope.

TABLE II
SENSITIVITY ANALYSIS OF THE SINGLE LOWER BOUND CONSTRAINT (H_{poa})

Lower Threshold H_{poa} (kWh/m ² /day)	R^2 Score	MSE
3.0	0.8921	0.005633
3.2	0.8818	0.006171
3.5	0.8457	0.008053
3.6	0.8233	0.009224
3.8	0.8850	0.006002
4.0	0.8918	0.005647
4.2	0.8553	0.007555

To identify the optimal activation threshold for the lower boundary constraint, a sensitivity analysis was performed by sweeping the H_{poa}^{lower} limit from 3.0 to 4.2 kWh/m²/day. The network was trained iteratively at each threshold using a fixed penalty factor ($\lambda = 0.01$) to observe how the strictness of the physical barrier influenced statistical accuracy.

The optimal thermodynamic comfort value was unequivocally identified at an H_{poa}^{lower} of **3.0 kWh/m²/day**. At this precise boundary, the model achieved peak variance explanation ($R^2 = \mathbf{0.8921}$) and minimum error (MSE = **0.005633**), successfully eradicating negative PR hallucinations without penalizing the natural variance of valid low-light generation.

Despite this optimization, the Single Bound PINN's peak performance remains below the dataset's theoretical potential. This vulnerability proves that rectifying only the lower operational envelope is insufficient, directly validating the necessity of deploying the full Double Bound architecture to manage upper-tier overclipping [13].

B. Full Thermodynamic Integration (The Double Bound PINN)

Having established that a partial boundary is insufficient to govern the full operational spectrum of the PV plant, the complete thermodynamic envelope was deployed. This final architecture, the Double Bound neural model, includes both the thresholds (H_{poa}^{lower} and H_{poa}^{upper}). By fully integrating both physical limits, the objective function takes its complete form.

$$\mathcal{L}_{total} = \text{MSE} + \lambda \cdot \left[\text{ReLU}(H_{poa}^{lower} - \hat{H}_{poa}) + \text{ReLU}(\hat{H}_{poa} - H_{poa}^{upper}) \right] \quad (13)$$

To identify the optimal physical corridor ($[H_{poa}^{lower}, H_{poa}^{upper}]$), a sensitivity analysis was conducted across five progressively narrower intervals using the previously calibrated penalty factor ($\lambda = 0.01$). The objective was to observe how the neural network's gradient descent adapted as the mathematical walls were tightened around the core operational data. The predictive outcomes are detailed in Table III.

TABLE III
SENSITIVITY ANALYSIS OF THE DOUBLE BOUND PHYSICAL CORRIDOR (H_{poa})

Physical Corridor H_{poa} (kWh/m ² /day)	R^2 Score	MSE
[2.5 - 5.5]	0.9043	0.004994
[3.0 - 5.0]	0.8877	0.005860
[3.2 - 4.8]	0.8935	0.005556
[3.5 - 4.5]	0.9111	0.004639
[3.8 - 4.2]	0.9118	0.004602

The empirical data demonstrates a highly effective regularization effect induced by the dual constraints. When evaluating the widest interval ($[2.5 - 5.5]$ kWh/m²/day), the model achieved a strong baseline R^2 of **0.9043**. However, because this envelope was highly permissive, the network still retained too much statistical freedom to extrapolate near the edges of physical impossibility during extreme weather anomalies.

The peak performance was recorded at the narrowest tested corridor of $[3.8 - 4.2]$ kWh/m²/day. At this highly constrained setting, the model achieved its maximum variance explanation ($R^2 = \mathbf{0.9118}$) and absolute minimum error margin (MSE = **0.004602**). By simultaneously guarding against low-irradiance startup errors and high-irradiance saturation hallucinations within a tight operational band, the Double Bound PINN successfully eradicated out-of-bounds predictions, proving that a rigidly calibrated thermodynamic envelope is the most effective method for ensuring stable regression in photovoltaic modeling.

To rigorously validate the proposed framework, as massive comparative analysis was executed across all six neural network architectures, testing 42 distinct configurations of the physics-informed loss function. The results, detailed in Table IV, map the intricate relationship between feature engineering, data distribution and thermodynamic boundary constraints. The data immediately reveals that feature selection dictates the model's fundamental capacity to learn.

The data immediately reveals that feature selection dictates the model's fundamental capacity to learn. Across all tests, the 6 calculated features (Architectures 1 and 4) severely underperformed, proving that manually pre-processing complex solar variables strips away the hidden, non-linear correlations that the network relies upon. Conversely, providing the network with the full spectrum of 16 combined features (Architectures 2 and 5) consistently unlocked the absolute highest predictive precision, granting the gradient descent algorithm the dimensional freedom required to map complex inverter states.

TABLE IV
COMPREHENSIVE PERFORMANCE ANALYSIS OF THE SIX PHYSICS-INFORMED ARCHITECTURES

Configuration (Architecture)	Physical Limit (H_{poa})	R^2 Score	MSE
Imbalanced Dataset (Natural Distribution)			
Architecture 1 (10 Raw Features)	3.5	0.8878	0.005856
	3.8	0.8783	0.006354
	4.0	0.9004	0.005200
	4.2	0.8885	0.005821
	[3.2 - 4.8]	0.8786	0.006340
	[3.5 - 4.5]	0.9113	0.004632
Architecture 2 (6 Calculated Features)	[3.8 - 4.2]	0.8791	0.006311
	3.5	0.7768	0.011652
	3.8	0.8248	0.009146
	4.0	0.8332	0.008708
	4.2	0.8228	0.009249
	[3.2 - 4.8]	0.7556	0.012757
Architecture 3 (16 Combined Features)	[3.5 - 4.5]	0.6420	0.018691
	[3.8 - 4.2]	0.7753	0.011728
	3.5	0.9396	0.003153
	3.8	0.9136	0.004508
	4.0	0.9305	0.003630
	4.2	0.9304	0.003635
Architecture 4 (10 Raw Features)	[3.2 - 4.8]	0.9233	0.004005
	[3.5 - 4.5]	0.9481	0.002710
	[3.8 - 4.2]	0.9020	0.005115
	3.5	0.9816	0.001311
	3.8	0.9815	0.001316
	4.0	0.9760	0.001712
Architecture 5 (6 Calculated Features)	4.2	0.9756	0.001735
	[3.2 - 4.8]	0.9816	0.001307
	[3.5 - 4.5]	0.9811	0.001345
	[3.8 - 4.2]	0.9749	0.001784
	3.5	0.9073	0.006603
	3.8	0.8898	0.007847
Architecture 6 (16 Combined Features)	4.0	0.8950	0.007478
	4.2	0.8848	0.008204
	[3.2 - 4.8]	0.9099	0.006418
	[3.5 - 4.5]	0.8892	0.007889
	[3.8 - 4.2]	0.8868	0.008064
	3.5	0.9883	0.000833
Architecture 7 (16 Combined Features)	3.8	0.9825	0.001247
	4.0	0.9833	0.001189
	4.2	0.9807	0.001378
	[3.2 - 4.8]	0.9861	0.000992
	[3.5 - 4.5]	0.9861	0.000992
	[3.8 - 4.2]	0.9835	0.001172
Balanced Dataset (Oversampled Minority Classes)			

Ultimately, this comprehensive sweep identifies Architecture 5 (16 Combined Features on a Balanced Dataset) as the definitive solution for photovoltaic monitoring. By achieving

an exceptional variance explanation of $R^2 = \mathbf{0.9883}$ with an MSE of **0.000833**, this specific configuration proves that maximum predictive robustness is attained only through a triad of methodologies: deep feature integration to capture nuance, synthetic data balancing to remove statistical bias, and the physics-informed Double Bound to guarantee absolute thermodynamic compliance.

VI. CONCLUSIONS

This study successfully developed and validated a Neural Model design to predict the PR of a PV power plant. By dynamically reverse-calculating the network's dimensionless output and applying a custom ReLU based penalty function, the MLP was successfully constrained.

The primary advantage of the double-bound model is the eradication of hallucinations. Standard MLP, even when optimized, remains mathematically blind to physical laws. By enforcing the H_{poa}^{lower} and H_{poa}^{upper} thresholds, the proposed architecture acts as an impenetrable mathematical safeguard. This transforms the neural network into a physically grounded monitoring tool.

Despite its robust performance, the proposed methodology exhibits several technical limitations. First of all, the architecture is highly sensitive to hyperparameter calibration because the penalty factor ($\lambda = 0.01$) and optimal physical corridor (e.g., the $[3.8 - 4.2]$ kWh/m²/day interval) were identified through extensive empirical searches.

So, due the fact this depends on the specific geographic coordinates of Iasi, Romania, the model is not immediately ready to use for other solar facilities. Finally, the model's peak accuracy heavily relies on the availability of 16 combined features, demanding a high-resolution, multi-sensor SCADA network that may not be available in older or lower-tier commercial solar parks.

The integration of physical constraints makes this model well-suited for industrial, real-world deployment. Its primary application lies in real-time fault detection and predictive maintenance within PV control rooms [12].

Moving forward, future research should focus on developing adaptive penalty mechanisms, rather than using a static λ factor and rigid H_{poa} boundaries. Ultimately, by bridging the gap between data-driven pattern recognition and fundamental thermodynamics, this study provides a scalable blueprint for creating safer and more reliable AI for renewable energy sector.

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